Pasabay

App Technical Specification

Version 2 — Second Iteration

USC Talamban Pilot

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1. Key Features Overview

The following features form the core of the Pasabay platform. Each is described in detail in subsequent sections.

Feature	Description
User Login & Sign Up	Email/password or Google OAuth restricted to .edu addresses
Identity Verification	School ID verification via AI image comparison; Driver's License via third-party service
Routing (OSRM)	OpenStreetMap Routing Machine for driver route calculation and passenger matching
Matching System	Spatial algorithm that pairs passengers within 300m of a driver's active route
Compliance Engine	Proprietary fare formula capping cost to actual fuel share plus a flat software fee
Money Handler	Xendit marketplace payment flow — passenger pays once, backend splits to driver and service fee
Driver Fuel Efficiency Spec	Driver-configurable fuel efficiency input validated against car make/model estimate
Rating System	Post-ride mutual rating; shadow-ban enforcement below a set threshold

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2. Application Pages & UI Flow

2.1 Splash Page

- Animated Pasabay icon displayed while the app loads.

2.2 Index Page

- Sign Up button
- Log In button
- Continue with Google (restricted to .edu email addresses only)

2.3 Sign Up Page

Inputs:

- School email (.edu only)
- Password
- Re-enter password

Actions:

- Terms of Service acceptance
- Sign Up button
- Continue with Google (.edu only)
- Link to Log In page

2.4 Log In Page

Inputs:

- School email
- Password

Actions:

- Log In button
- Continue with Google (.edu only)
- Forgot Password
- Link to Sign Up page

2.5 School ID Verification Page

Triggered immediately after account creation. The user **must** verify their USC identity before proceeding.

UI Elements:

- Vertical camera frame for lining up school ID
- Pop-up tips: well-lit room, avoid glare
- Liveness check (confirm user is physically present)
- Manual input fallback if camera capture fails

Note: *Backend uses EfficientNet-B0 image similarity against a stored USC ID reference template. Hosted locally for Phase 1.*

2.6 Driver Verification Page

After School ID verification, the user is prompted to optionally register as a driver. This can be changed later in Settings.

If the user opts in as a driver:

- Vertical camera frame to scan Driver's License
- Privacy agreement and explanation of why the license is needed
- Pop-up tips: well-lit room, avoid glare
- Liveness check
- Manual input fallback

Vehicle Detail Page (shown after license scan):

- License plate number
- Car color
- Car make and model
- Car seat capacity
- Car year (used to validate fuel efficiency input)

Note: *Phase 1 uses **Persona** for Driver's License verification. Migration to **Shufti Pro** is planned for Phase 2/3 due to stronger Philippine provincial coverage.*

2.7 Homepage Stack

The homepage differs based on whether the logged-in user is a Driver or Passenger.

Driver View

- Map interface modeled on Waze UI with real-time navigation
- Driver inputs their destination to activate their route
- When a match is offered by the system, a minimal pop-up is shown with Accept / Deny options
  - If accepted: driver is shown the passenger pickup point and begins a 1-minute wait timer
  - If the timer expires and the passenger has not arrived: a "Passenger No-Show" button appears to cancel the match
  - If denied: driver continues on their original route

Passenger View

- Map interface with a destination input field
- On input confirmation, the matching system is triggered
- Pre-match pop-up shows: Terms & Conditions, expected passenger behavior, and the estimated fare. (Will show as well how much is deducted if more people sit in)
- Matching animation shown while the system searches for a driver
- On successful match: passenger is shown their standby location and a walking path to the pickup point
- ETA of driver to pickup point is shown (updated every 30 seconds — lightweight, not full real-time tracking)

Note: *Full real-time driver location tracking is deferred to Phase 2. Phase 1 uses a 30-second ETA refresh as a practical substitute.*

Previous Routes Tab

- Displays history of completed and canceled rides
- Tapping a route shows a pop-up with full trip details

Profile & Settings Tab

- User profile showing Driver or Passenger status
- Option for passengers to register as a driver
- Payment Methods management
- Emergency Contact
- Logout
- Delete Account

3. Identity Verification

Identity verification is split into two separate flows: School ID verification (required for all users) and Driver's License verification (required only for drivers).

3.1 School ID Verification Flow

Step 1 — Camera Capture:

- App opens device camera in a rectangular frame
- User aligns their USC ID within the frame
- Photo is taken automatically when an ID is detected, or manually via shutter button

Step 2 — AI Similarity Check (Backend):

- Uploaded image is compared against stored USC ID reference templates using EfficientNet-B0 feature extraction
- Cosine similarity score is calculated; a score above the defined threshold constitutes a pass
- Scores in the uncertain middle range (0.75–0.85) are flagged for manual review

Step 3 — Result:

- Pass: student name and ID number are extracted and stored; user proceeds to Driver Verification prompt
- Fail: failure screen is shown with option to retry or use manual input fallback

Note: *The AI model is hosted on a local server for Phase 1. No third-party cost per verification. Migration to cloud hosting is the natural Phase 2 trigger.*

3.2 Driver's License Verification Flow

Step 1 — Camera Capture:

- App opens device camera; user aligns their Philippine Driver's License
- Photo taken automatically or via shutter button

Step 2 — Third-Party Verification:

- Image is sent to Persona (Phase 1) for document authenticity check and OCR extraction
- Liveness detection confirms the person presenting the license is real

Step 3 — Result:

- Pass: license data stored against user profile; user proceeds to Vehicle Details page

- Fail: failure screen shown with retry or manual input fallback

3.3 Verification Service Roadmap

Service	Type	PH License	Liveness	Phase
Persona	Full IDV	Yes (global)	Yes	Phase 1
Shufti Pro	Full IDV	Yes (all 81 PH provinces)	Yes	Phase 2/3
eKYC Pro	OCR only	Yes (PH-specific)	Separate	Not recommended
Onfido	Full IDV	Yes	Yes	Enterprise only

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4. Matching System

The matching system pairs passengers with drivers sharing the same travel corridor. It is primarily triggered by the passenger's destination input.

4.1 Core Concepts

When a driver sets their destination, OSRM returns their trip as a Polyline — a sequence of GPS coordinates representing the route. Passenger pickup and drop-off locations are treated as Points. The system checks whether these Points fall within a 300-meter buffer zone around the Driver's Polyline.

4.2 Matching Algorithm Steps

The backend runs the following sequence for every new passenger request against all active driver routes:

Check	Description
Proximity — Origin	Calculate perpendicular distance from the passenger's starting point to the nearest segment of the driver's polyline. Must be ≤ 300m.
Proximity — Destination	Calculate distance from the passenger's intended drop-off to the nearest segment of the polyline. Must be ≤ 300m.
Directionality Check	Project both passenger points onto the driver's route and verify the pickup index occurs before the drop-off index. Prevents matching with drivers going in the opposite direction on the same road.

Snap-to-Route Drop-off	The exact coordinate on the driver's route closest to the passenger's true destination becomes the official drop-off point. The driver does not detour.
Time Synchronization	Verify the driver's real-time ETA to the proposed pickup coordinate. If the driver has already passed the pickup point, the match is invalid.

4.3 Edge Cases

- Divided Highway Problem: A passenger may be within 300m straight-line distance but on the opposite side of a divided highway. Street heading and direction must be factored in, not just radius.
- Time Synchronization: Spatial proximity alone is not sufficient. The system must validate that the driver has not yet passed the passenger's pickup point.
- Walking UI: Since passengers walk to the pickup spot and from the drop-off spot, the UI must clearly visualize both walking paths before the match is confirmed.

4.4 Infrastructure Notes

- Geospatial database: PostGIS recommended
- Point-to-polyline projection algorithms required
- 300m buffer zone visualization in the UI for testing/debugging

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5. Compliance Engine

The Compliance Engine ensures every fare is mathematically capped to the passenger's proportional fuel cost, keeping all rides legally classified as non-commercial cost-sharing under LTFRB regulations.

5.1 Fare Formula

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Passenger Fare = [ (Passenger Distance / Sum of All Passenger Distances) × (Total Route Distance × Fuel Price / Fuel Efficiency) ] + Matching Fee
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Variable	Definition
Passenger Distance	The distance covered by this specific passenger on the trip.
Sum of All Passenger Distances	The combined distances of all passengers on the same driver trip.

Total Route Distance	The full distance from the driver's origin to the last passenger's drop-off.
Fuel Price	Current DOE weekly pump price (updated via DOE feed).
Fuel Efficiency	Default: 20 km/L. Overridden by driver's validated custom input (see Section 7).
Matching Fee	Flat PHP 5–10 charged as a separate software service fee. Never touches transport money.

Note: *The matching fee is collected separately via Xendit's Platform Fee feature, creating a clean audit trail for LTFRB compliance.*

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6. Money Handler

The payment architecture uses a marketplace disbursement model via Xendit. The passenger pays a single amount; the backend automatically splits it before disbursing.

6.1 Payment Flow

Step	Action
1	Passenger pays full amount (fare + matching fee) via GCash or Maya through Xendit checkout.
2	Xendit holds the full payment.
3	Backend triggers split: driver's fuel share is disbursed to driver's GCash/Maya wallet; matching fee is routed to Pasabay's merchant account.
4	Transaction record is stored. Both parties receive a payment confirmation.

6.2 Why This Architecture

- Passenger pays only once — no two-step or split-payment UX burden on the user
- Matching fee is provably separate from transport money — key to the LTFRB non-commercial classification
- Xendit's XenPlatform feature handles the split natively via their Platform Fee API
- Xendit disburses to 94+ banks and 13 Philippine e-wallets including GCash and Maya

6.3 No-Show & Refund Logic

- If a driver triggers "Passenger No-Show" after the 1-minute window: payment is voided/reversed before disbursal
- If a match is canceled before payment is submitted: no transaction occurs
- Physical cash fallback: documented as a known edge case for Phase 1; full virtual-only enforcement is Phase 2

Note: *Since Pasabay briefly holds passenger funds before disbursement, consult legal counsel on BSP e-money aggregator regulations before scaling beyond Phase 1. Routing through Xendit (BSP-supervised) likely provides coverage as their merchant.*

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7. Driver Fuel Efficiency Spec

Drivers can override the default fuel efficiency (20 km/L) in the Compliance Engine formula with their vehicle's actual efficiency. This is validated against an estimated range for their registered car make, model, and year.

7.1 Validation Logic

- Driver inputs custom fuel efficiency (km/L) during vehicle setup or in Settings
- Backend retrieves estimated efficiency range for the car's make, model, and year from a reference dataset
- If the driver's input falls within an acceptable tolerance of the estimated range, it is approved
- If the input is outside the tolerance (e.g., unrealistically high to manipulate fares), it is rejected and the default is applied

7.2 Helper Calculator

An optional on-screen calculator helps drivers estimate their fuel efficiency:

- Input: Distance traveled (km)
- Input: Fuel consumed (liters)
- Output: Calculated km/L

7.3 Vehicle Details Required

- License plate number (for identity binding)
- Car make (e.g., Toyota, Honda)
- Car model (e.g., Vios, City)
- Car year (used for efficiency estimation lookup)

- Car color (shown to passengers for identification)

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8. Rating System

The rating system maintains trust and safety on the platform by enabling mutual post-ride feedback with automated enforcement.

8.1 Mechanics

- After every completed ride, both the driver and the passenger are prompted to rate each other
- Ratings are on a standard scale (e.g., 1–5 stars)
- Both parties must rate before the next ride can be initiated (or a timeout dismisses the prompt)

8.2 Enforcement

- If a user's aggregate rating drops below a defined threshold, they are shadow-banned from the platform
- Shadow-ban: the user can still log in and use the app, but they are silently excluded from the matching pool
- Shadow-ban threshold and review/appeal process to be defined before Phase 1 launch

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9. Connectivity Handling

Pasabay does not support offline operation — the core matching, routing, and payment flows require a live connection. The following graceful degradation behaviors are implemented for poor or lost connectivity.

Scenario	Behavior
Connection lost during matching	Matching animation pauses; "No Connection" banner displayed. Resumes automatically when connection is restored.
Connection lost mid-ride	GPS caches last known location briefly. A banner warns that live tracking is temporarily unavailable.
Connection lost at payment	Payment action is blocked with a clear error message. Transaction is not submitted in a partial state.

Slow connection (not lost)	Request timeouts with automatic retries prevent the UI from freezing indefinitely.
Map rendering	Map tiles are cached locally so the map base layer renders even when live data is slow.